Marula School Manager: role relationships and use cases

Review of abcSchool_Manager.html, covering the admin, teacher, student and parent roles, the gaps found, and proposed new use cases.

How the roles relate

- The Administrator creates and manages every record.
- A Teacher teaches one or more Classes (each teacher has one subject).
- A Class has many Students and one class teacher.
- A Student has a Parent as guardian (`student.guardian`). A parent can have many students, but a student has only one guardian today.
- Fees, marks and attendance all hang off the student.
- Parent and teacher have no direct channel today.

What each role can do today

Module	Admin	Teacher	Student	Parent
Students	Edit	View all students	–	–
Teachers, Parents, Settings	Edit	–	–	–
Classes, Subjects	Edit	View	–	–
Attendance	Edit	Edit (own classes)	View own	View children
Exams and results	Edit	Edit	View own	View children
Assignments, Timetable	Edit	Edit assignments	View	View
Fees	Edit, record payments	–	View own	View children

Module	Admin	Teacher	Student	Parent
Library	Edit	View	View	–
Reports	Edit	View	–	–

Problems found

1. **No real identity.** The role is a demo dropdown, and the identities are hardcoded (student is always `S1001`, parent `P01`, teacher `T01`). Anyone can switch to Administrator. A real version needs login, with the role and linked ID coming from the account and checked on the server, not just hidden in the UI.
2. **Results CSV leak.** "Export CSV" on the Results page exports the whole class's marks, even for a student or parent, because it filters by class instead of by their own children.
3. **Teachers see fee data.** The student profile and the Reports Fees tab both show balances. Teachers also see every student in the school, because `scoped()` only filters for student and parent roles.
4. **Teacher write access is too wide.** A teacher can edit marks for all six subjects, not just their own. They can also add, edit or delete exams and assignments for any class.
5. **Students and parents see other classes' exams.** A 7A student sees the 9B science practical on the dashboard.
6. **The teacher, class and subject link is too thin.** The timetable takes the first teacher with that subject and ignores class assignments. T01 and T07 both teach Maths, so it can name a teacher who never teaches that class. Marks are not linked to exams, and assignments have no owner or submissions.
7. **Orphaned records on delete.**
   - Deleting a student leaves payments and attendance behind.
   - Deleting a teacher leaves `class.teacher` dangling.
   - Deleting a parent leaves students pointing at nothing.
   - Deleting a fee leaves payment rows, which then show "—" in Recent payments.

Model changes that make the use cases work

- **User account:** login, role, and the linked teacher, student or parent ID.
- **Parent and student link:** many-to-many, with primary and secondary guardians.

- **Teaching assignment:** teacher, class and subject together.
- **New records:** submissions, messages, announcements, leave requests and an audit log.

Cross-role flows

- **Absence:** the parent reports it, the teacher sees it on the attendance sheet and marks Excused, and the parent gets a notification.
- **Assignment:** the teacher posts it, the student submits, the teacher grades, and parent and student see the result.
- **Results:** the teacher enters marks for their subject, the admin reviews and publishes, then student and parent can view and download the report card.
- **Fees:** the admin bills by class and term, the parent pays, the admin reconciles and issues a receipt, and reminders go out for overdue balances.

Use cases per role

Items marked (new) are proposed additions.

Administrator

- Existing: manage all records, record payments, run reports, change settings.
- (new) Create accounts and link parents to children.
- (new) Assign teachers to class and subject, and arrange cover when one is on leave.
- (new) Promote classes at year end.
- (new) Set fee structures per class.
- (new) Publish or lock results.
- (new) Post school-wide announcements.
- (new) View the audit log.

Teacher

- Existing: take attendance, enter marks, set exams and assignments, view timetable and library.
- (new) See a class roster with guardian contacts, without fee data.
- (new) Enter marks only for their own subject, per exam.

- (new) Grade assignment submissions.
- (new) Write report-card comments.
- (new) Message parents and post class announcements.
- (new) Request leave.

Student

- Existing: view own attendance, exams, assignments, timetable, fees and library.
- (new) Submit assignments.
- (new) See grade trends per subject.
- (new) Download report cards.
- (new) See books borrowed and due dates.
- (new) Read announcements.

Parent

- Existing: view children's attendance, results, fees, exams and timetable.
- (new) Switch between children from one login.
- (new) Report an absence in advance.
- (new) Pay fees online and download receipts.
- (new) Message teachers and book parent-teacher meetings.
- (new) Get alerts for absences, low grades and fees due.
- (new) Give consent for trips.